

MATH I – Week 2

Practical session problems

Exercise 1: Relations and functions

Determine whether each relation defines y as a function of x . Explain your answer.

a. $y = 3x - 2,$

b. $x^2 + y^2 = 9,$

Where appropriate, solve for y and apply the vertical-line test.

Exercise 1: Relations and functions

Solution

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a. $y = 3x - 2$,

b. $x^2 + y^2 = 9$,

Where appropriate, solve for y and apply the vertical-line test.

a. $y = 3x - 2$ is a function: every x produces exactly one y .

b.

$$x^2 + y^2 = 9 \implies y = \pm\sqrt{9 - x^2}.$$

It is not a function of x on $[-3, 3]$, because most values of x produce two values of y .

Exercise 2: Function notation

Let

$$f(x) = 2x^2 - 3x + 1.$$

Calculate and simplify:

$$f(0), \quad f(-2), \quad f(a), \quad f(a+h),$$

$$\frac{f(a+h) - f(a)}{h}, \quad h \neq 0.$$

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For $f(x) = 2x^2 - 3x + 1$,

and

Expanding,

Therefore,

and, for $h \neq 0$,

Solution

$$f(0) = 1,$$

$$f(-2) = 2(-2)^2 - 3(-2) + 1 = 15,$$

$$f(a) = 2a^2 - 3a + 1,$$

$$f(a+h) = 2(a+h)^2 - 3(a+h) + 1.$$

$$f(a+h) = 2a^2 + 4ah + 2h^2 - 3a - 3h + 1.$$

$$f(a+h) - f(a) = 4ah + 2h^2 - 3h,$$

$$\frac{f(a+h) - f(a)}{h} = 4a + 2h - 3.$$

Exercise 3: Domain

Find the natural domain of each real-valued function:

a. $f(x) = \frac{x+1}{x^2-9},$

b. $g(x) = \sqrt{5-2x},$

Express each domain using interval notation.

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Solution

Problem 3: Domains

a.

$$f(x) = \frac{x+1}{x^2-9}.$$

The denominator cannot be zero:

$$x \neq -3, \quad x \neq 3.$$

Thus,

$$D_f = (-\infty, -3) \cup (-3, 3) \cup (3, \infty).$$

b.

$$g(x) = \sqrt{5-2x}.$$

We require $5-2x \geq 0$, so $x \leq 5/2$. Therefore,

$$D_g = (-\infty, 5/2].$$

Exercise 4: Composite functions

Let

$$f(x) = 2x - 3, \quad g(x) = x^2 + 1.$$

Find:

$$(f \circ g)(x), \quad (g \circ f)(x),$$

Are $f \circ g$ and $g \circ f$ equal? Explain why function composition is generally not commutative.

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Solution

Problem 4: Composite functions

$$(f \circ g)(x) = f(x^2 + 1) = 2x^2 - 1,$$

$$(g \circ f)(x) = g(2x - 3) = (2x - 3)^2 + 1 = 4x^2 - 12x + 10,$$

In general,

$$f \circ g \neq g \circ f.$$

Therefore, function composition is not commutative.

Exercise 5: Symmetry

Determine whether each function is even, odd or neither:

$$|x| + 1, \quad \frac{x}{x^2 + 1}.$$

Verify your conclusions algebraically and graphically.

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Problem 5: Symmetry

Use the tests $f(-x) = f(x)$ for even functions and $f(-x) = -f(x)$ for odd functions.

a) Even, b) Odd

Exercise 6: Identify the function class ¶

Classify each function as linear, polynomial, rational, algebraic or transcendental. A function may belong to more than one category.

$$\frac{x + 1}{x^2 + 4},$$

$$\sqrt{x^2 + 1}$$

Solution

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$$\sqrt{x^2+1}$$

Problem 6: Function classes

Function	Classification
$\frac{x+1}{x^2+4}$	Rational and algebraic
$\sqrt{x^2+1}$	Algebraic

Exercise 7: Linear engineering model

The temperature inside an industrial chamber increases linearly from 20°C at $t = 0$ to 80°C at $t = 15$ minutes.

1. Find a function $T(t)$ describing the temperature.
2. Interpret its slope and vertical intercept.
3. Find $T(8)$.
4. At what time does the chamber reach 60°C ?
5. State a physically reasonable domain.

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Solution

Problem 7: Linear temperature model

The slope is

$$m = \frac{80 - 20}{15 - 0} = 4^{\circ}\text{C}/\text{min}.$$

Therefore,

$$T(t) = 20 + 4t.$$

At $t = 8$,

$$T(8) = 20 + 4(8) = 52^{\circ}\text{C}.$$

For $T = 60$,

$$60 = 20 + 4t \implies t = 10 \text{ minutes}.$$

For the stated heating period, a reasonable domain is

$$0 \leq t \leq 15.$$

The intercept is the initial temperature, and the slope is the heating rate.

Exercise 8: Polynomial behaviour

For

$$p(x) = x^3 - 4x$$

determine:

- domain;
- intercepts;
- symmetry, if any;
- long-run or end behaviour;
- locations where the graph is undefined.

Exercise 8: Polynomial behaviour

For

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Solution

determine:

- domain;
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- symmetry, if any;
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Problem 8: Polynomial behaviour

For

$$p(x) = x^3 - 4x = x(x - 2)(x + 2) :$$

- Domain: $\mathbb{R}$
- x -intercepts: $(-2, 0)$, $(0, 0)$, $(2, 0)$
- y -intercept: $(0, 0)$
- Symmetry: odd
- As $x \rightarrow \infty$, $p(x) \rightarrow \infty$
- As $x \rightarrow -\infty$, $p(x) \rightarrow -\infty$

Exercise 9: Piecewise electricity tariff

An electricity provider charges according to

$$C(E) = \begin{cases} 0.10E, & 0 \leq E \leq 200, \\ 20 + 0.15(E - 200), & E > 200, \end{cases}$$

where E is monthly energy consumption in kWh and $C(E)$ is the cost.

1. Find $C(150)$, $C(200)$ and $C(350)$.
2. Sketch the graph.
3. Is the cost model continuous at $E = 200$?
4. Explain the meaning of each slope.
5. Suggest one limitation of the model.

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Problem 9: Piecewise Electricity tariff

$$C(150) = 0.10(150) = 15,$$

$$C(200) = 0.10(200) = 20,$$

$$C(350) = 20 + 0.15(350 - 200) = 42.50.$$

At $E = 200$, both formulas produce 20, so the model is continuous there.

The first slope represents a cost of 0.10 currency units per kWh. The second represents 0.15 currency units per additional kWh.

A limitation is that the model excludes fixed charges, taxes and changes in tariff over time.

Exercise 10: Finding inverse functions

Find the inverse of the following function and state the domain of the function and its inverse:

$$f(x) = 4x - 7,$$

Verify your answer using composition:

$$f(f^{-1}(x)) = x.$$

Exercise 10: Finding inverse functions

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Solution

Problem 10: Inverses (finding inverse functions)

a.

$$y = 4x - 7 \implies x = \frac{y + 7}{4}.$$

Thus,

$$f^{-1}(x) = \frac{x + 7}{4}.$$

Both domains are $\mathbb{R}$.

Exercise 11: Exponential growth and decay

For each model, decide whether it represents growth or decay and identify its initial value and growth or decay factor:

$$A(t) = 500(1.08)^t,$$

$$B(t) = 1200(0.85)^t,$$

$$C(t) = 40e^{0.03t},$$

$$D(t) = 200e^{-0.4t}.$$

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Problem 11: Growth and decay

Model	Type	Initial value	Factor/rate
$500(1.08)^t$	Growth	500	Factor 1.08 , or 8% growth per unit time
$1200(0.85)^t$	Decay	1200	Factor 0.85 , or 15% decay per unit time
$40e^{0.03t}$	Growth	40	Continuous rate 0.03
$200e^{-0.4t}$	Decay	200	Continuous decay rate 0.4

Exercise 12: Solve exponential equations

Solve:

$$2^{x+1} = 16,$$

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Solution

Solve:

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Problem 12: Exponential equations

a.

$$2^{x+1} = 16 = 2^4 \implies x = 3.$$

Exercise 13: Battery discharge

The stored energy in a battery is modeled by

$$E(t) = E_0 e^{-0.025t},$$

where t is measured in hours.

1. What fraction of the initial energy remains after 10 hours?
2. When will 50% of the initial energy remain?
3. When will 10% remain?
4. Explain why the model never predicts exactly zero energy.

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Solution

Problem 13: Battery discharge

$$E(t) = E_0 e^{-0.025t}.$$

After 10 hours,

$$\frac{E(10)}{E_0} = e^{-0.25} \approx 0.7788.$$

Thus approximately 77.9% remains.

For 50%,

$$e^{-0.025t} = 0.5,$$

so

$$t = \frac{\ln 2}{0.025} \approx 27.73 \text{ h.}$$

For 10%,

$$e^{-0.025t} = 0.1,$$

so

$$t = \frac{\ln 10}{0.025} \approx 92.10 \text{ h.}$$

The exponential is always positive for finite t , so the model approaches zero asymptotically but never reaches it exactly.

Exercise 14 (if time allows): Population growth

A bacterial population is modeled by

$$P(t) = 800e^{0.35t},$$

where t is measured in hours.

1. Find the initial population.
2. Find the population after 5 hours.
3. Determine the doubling time.
4. Explain whether the model can remain realistic indefinitely.

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Solution

Problem 14: Population growth

$$P(t) = 800e^{0.35t}.$$

The initial population is

$$P(0) = 800.$$

After five hours,

$$P(5) = 800e^{1.75} \approx 4604.$$

For the doubling time,

$$1600 = 800e^{0.35t},$$

so

$$t = \frac{\ln 2}{0.35} \approx 1.98 \text{ h.}$$

The model cannot remain realistic indefinitely because resources, space and nutrients are limited.